

# Impact of livestock and settlement on the large mammalian wildlife of Bale Mountains National Park, southern Ethiopia

Philip A. Stephens<sup>a,\*</sup>, Candy A. d'Sa<sup>b,1</sup>,  
Claudio Sillero-Zubiri<sup>b</sup>, Nigel Leader-Williams<sup>a</sup>

<sup>a</sup>Durrell Institute of Conservation and Ecology, Canterbury, Kent, CT2 7NJ, UK

<sup>b</sup>Wildlife Conservation Research Unit, Department of Zoology, University of Oxford, Oxford OX1 3PS, UK

Received 6 September 2000; received in revised form 14 November 2000; accepted 7 December 2000

## Abstract

Large mammals, both wild and domestic, were censused in four study areas in Bale Mountains National Park (BMNP), southern Ethiopia, from April until August 1997, using established road counts, horse-back counts and systematic transects, a new approach to censusing large mammals in BMNP. Data collected since 1983 were also examined to indicate trends in mammalian abundance. Civil unrest following the government changeover in 1991 disrupted management of BMNP, leading to an increase in human settlement and livestock densities. The effect of these factors on BMNP's wildlife is assessed and discussed. Most large mammalian wildlife has declined since 1983 but signs of recovery for some species in recent years are positive. The fates of two charismatic endemics are emphasised. BMNP's current population of mountain nyala (*Tragelaphus buxtoni*, an endangered antelope) was estimated to be between 1100 and 1300. This is lower than estimates for the late 1980s, but shows an increase from very low levels recorded following social turmoil during the government changeover in 1991. Data on the Ethiopian wolf (*Canis simensis*, a critically endangered canid) also indicate positive trends in abundance, following severe declines between 1989 and 1992 due to rabies epizootics. Action to reduce human utilisation of parts of BMNP is urgently required to prevent further degradation of the environment and to safeguard the future of both the mountain nyala and Ethiopian wolf. © 2001 Elsevier Science Ltd. All rights reserved.

**Keywords:** Bale Mountains; Census; Distance sampling; *Canis simensis*; *Tragelaphus buxtoni*

## 1. Introduction

### 1.1. Background

Although protected areas are one of conservation's oldest devices and remain a cornerstone of conservation policy (Wynne, 1998), such areas are continually under threat from growing human populations. This is particularly the case in the developing world, where growing

populations threaten protected areas directly by encroachment of settlements but also through the pressure on all land to become economically productive, for example, as areas for timber production or livestock grazing. It is important for conservationists to have detailed information on the type of impacts that result from expanding human populations in areas of high biodiversity. This, in turn, relies on the use of long-term data sets, to assess trends in populations of fauna or flora within an anthropogenic context.

Ethiopia is remarkable for its endemic fauna and flora, which arise from the vast extent and isolation of its highlands within the Afrotropical region (Yalden, 1983). Much of the diversity of Ethiopian wildlife occurs at high altitudes and in 1974 the Bale Mountains National Park (BMNP) was designated, primarily to safeguard populations of two of the country's most

\* Corresponding author. Present address: School of Biological Sciences, University of East Anglia, Norwich NR4 7TJ, UK. Tel.: +44-01603-593132.

E-mail address: philip.stephens@uea.ac.uk (P.A. Stephens).

<sup>1</sup> Present address: School of Biological Sciences, University of Southampton, Southampton, SO16 7PX, UK.

charismatic mammalian endemics — the mountain nyala (*Tragelaphus buxtoni*) and the Ethiopian wolf (*Canis simensis*) — but also to protect part of the largest tract of Afroalpine habitats on the continent (Brown, 1969). At that time, the area was largely uninhabited but under the socialist-military government of 1974–1991, increasing investment in mechanised state farms in the lowlands left little room for pastoralists, except at higher altitudes. Since then BMNP has, like so many of Africa's protected areas, been under increasing pressure from an ever-growing human population.

A protracted civil war in Ethiopia was followed in May 1991 by the replacement of the socialist military regime by the current democratic government. During a brief period of unrest associated with the transition between governments, firearms became readily available and wildlife became the target for expression of resentment over restrictions and prevention of activities such as settlement, grazing, cultivation and collection of firewood. Many mountain nyala and some bohor reedbuck (*Redunca redunca*) and Menelik's bushbuck (*Tragelaphus scriptus menelikii*) were speared or shot (Woldegebriel, 1996). Ethiopian wolves were also the target of persecution; at least six killings were recorded in late 1991 (Sillero-Zubiri and Gottelli, 1993) and many more may have occurred in the immediate aftermath of the civil unrest. In addition to the direct persecution of wildlife, previously cleared areas were resettled, resulting in considerable destruction of habitat.

Despite these setbacks, the body of scientific information compiled since 1983 by BMNP staff and other researchers remains considerable. As a result, BMNP is one of the few areas in Ethiopia where the long-term effects of anthropogenic influences on mammalian wildlife may be assessed. This study, conducted from April to August 1997, aimed to reinstate routine censusing of the area and to collate data from previous periods of censusing in BMNP. These data were used to study the relationship between human settlement, livestock grazing and trends in populations of the large mammals of BMNP.

## 2. Methods

### 2.1. Study area

BMNP is located between 6°29' and 7°10'N, and between 39°28' and 39°58'E. The area of BMNP as designated in 1974 is 2200 km<sup>2</sup>, varying in altitude from 1500 to 4377 m above sea level (a.s.l.; Hillman, 1986). Detailed descriptions of the physical characteristics, fauna and flora of BMNP may be found elsewhere (Hillman, 1986; Malcolm and Sillero-Zubiri, 1997). Briefly, the study area is subject to a four month dry season from November to February, with hot days and

cold nights, and a more temperate wet season from March to October. Rainfall peaks in April and September (Sillero-Zubiri et al., 1998). The vegetation is largely of the Afroalpine and subalpine types (Malcolm and Sillero-Zubiri, 1997). BMNP harbours 24% of Ethiopia's mammalian species (66 of 277), including 55% of its endemics (17 of 31), and 30% of the bird species (262 of 861), including 57% of endemics (16 of 28), as well as many amphibian endemics (Hillman, 1993). Until relatively recently the Bale Mountains were very sparsely populated by humans. In 1958 the area was virtually uninhabited but analysis of aerial photography from both 1972 and 1984 showed that the human population of BMNP had been increasing since its proclamation (Hillman, 1986).

### 2.2. Study sites

This study focused on four sites in northern and central BMNP (see map, Fig. 1), representative of all the more open Afroalpine habitats.

#### 2.2.1. Gaysay

This area at about 3000 m a.s.l., forms the northern projection of BMNP and comprises open and swamp grasslands, mixed with *Artemisia* and *Helichrysum* spp. bush. The area is regularly patrolled but, nonetheless, is under growing pressure from domestic livestock. Gaysay is the most important area in BMNP for mountain nyala, bushbuck and reedbuck. Based on an index of diurnal rodent prey biomass, the montane grassland zone within Gaysay has been classified as good habitat for Ethiopian wolves (Gottelli and Sillero-Zubiri, 1992).

#### 2.2.2. Gojera

Gojera was thought to be an important corridor for wildlife (particularly mountain nyala) between the northern Gaysay area and the ericaceous heath along and around the eastern boundary of BMNP (Hillman, 1986). Between 1986 and 1991, this area was extended northward but was rapidly resettled following the change of government in 1991. Gojera slopes gently from 3600 m a.s.l. in the south to 3200 m a.s.l. in the north. Vegetation includes meadows of red-hot poker (*Kniphofia foliosa*) on the flatter ground, with heavily grazed areas of poorer drainage. *Artemisia afra* and *Helichrysum* spp. bush is mixed with ericaceous vegetation in various stages of development on the rockier ground and steep slopes.

#### 2.2.3. Upper Web Valley

This broad, flat valley (c. 3500 m a.s.l.) to the southwest of Gaysay consists of large areas of swamp and marsh through which the Web River meanders. The valley is walled to the west by a ridge of peaks and to

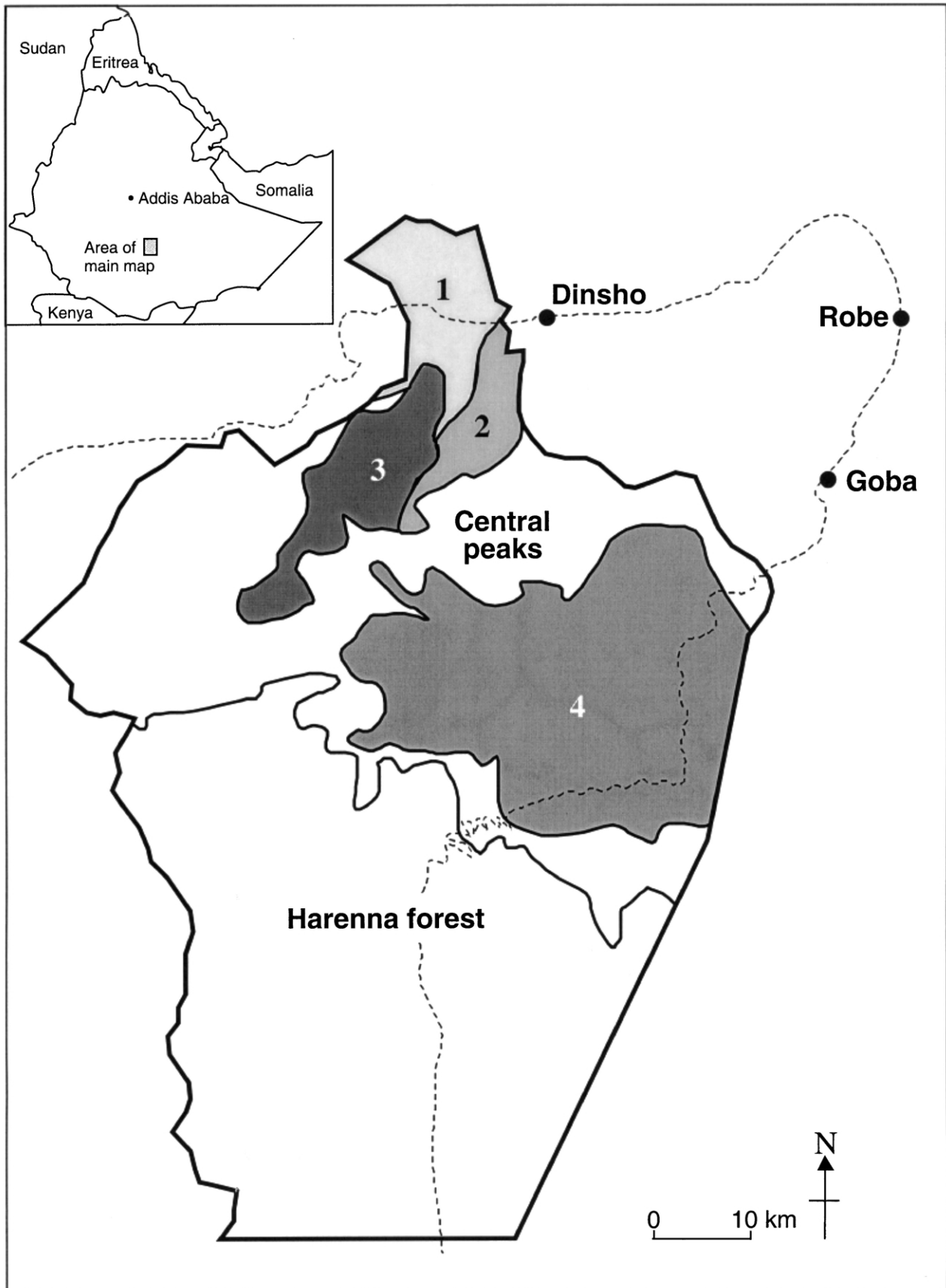


Fig. 1. Map of study sites in the Bale Mountains National Park. (1) Gaysay, (2) Gojera, (3) Upper Web Valley, (4) Sanetti Plateau. Main towns (filled circles) and road (dashed line) are also shown.

the south by lava cliffs up to 80 m high. The area is in the rain-shadow of the main mountain massif and the vegetation is largely of the Afroalpine type, with *Alchemilla* pasture dominating. Low ridges support grassy pasture and *Helichrysum* spp. shrubs. Steeper slopes are covered with *Helichrysum*/*Artemisia* scrub, with fennel (*Ferula communis*) and occasional stands of giant lobelia (*Lobelia rhynchopetalum*). The valley walls support thick, 1–2 m tall ericaceous vegetation (*Erica* and *Philippia* spp.). A distinctive feature of much of the valley floor is the pattern of low mounds, 10–20m in diameter, apparently formed by the fossorial activity of the giant mole rat (*Tachyoryctes macrocephalus*; Sillero-Zubiri et al., 1995b). The Upper Web Valley harbours the highest biomass of rodents of any area studied in BMNP (Sillero-Zubiri et al., 1995a) and has been classified as optimal habitat for Ethiopian wolves, which feed primarily on diurnal rodents (Gottelli and Sillero-Zubiri, 1992). Human settlement had been consistently recorded in the Upper Web Valley since before BMNP's proclamation (Brown, 1969; Hillman, 1986). Several settlements are located around the edges of the valley, and livestock concentrations are high (Sillero-Zubiri, 1994).

#### 2.2.4. Sanetti Plateau

At an altitude of over 4000 m a.s.l., this plateau is a major feature of east-central BMNP. Sanetti comprises three major habitat zones: ericaceous moorland is marginal habitat for Ethiopian wolves (Gottelli and Sillero-Zubiri, 1992), but is used by mountain nyala, bushbuck, grey duiker (*Sylvicapra grimmia*) and klipspringer (*Oreotragus oreotragus*) (Hillman, 1986); Afroalpine grassland is optimal wolf habitat, with limited other wildlife; *Helichrysum* scrub, dominated by *H. splendens* and *H. citrispinum* is also good habitat for wolves (Gottelli and Sillero-Zubiri, 1992). Settlement around the Sanetti Plateau is limited and numbers of livestock

have, in the past, been low by comparison to the Upper Web Valley (Hillman, 1986).

#### 2.3. Collation of existing data

All available census data were collated from published work, unpublished reports and BMNP records. Data spanned the period from 1983 to 1997 and were principally from road counts but also included regular circuit counts from horseback, and some transect results. The methods used varied between different areas of BMNP (Table 1). Census data on all large mammals were collated, where large mammals included all species with a body mass of over 10 kg (thus excluding rodents, lagomorphs and smaller carnivores which are relatively difficult either to count or to observe). Where domestic animals were recorded, these were classified as cattle, transport (horses, donkeys or mules), small stock (sheep or goats) and domestic dogs.

#### 2.4. Collection of census data

Census data were collected between late April and August 1997, where possible following existing census techniques and routes in each study area for ease of comparison (Table 1). Three census techniques were used: (1) bimonthly horseback circuits in the Web Valley, in which all large mammals were counted, using an established route that maximised visibility of the area (Sillero-Zubiri, 1994). The circuit census conducted in this area has employed a consistent route since its inception in 1988 and it was therefore possible to compare absolute numbers of animals sighted during these censuses over the years; (2) road counts were conducted from a vehicle in Gaysay and on the Sanetti Plateau, and from horseback in Gojera. All large mammal sightings were recorded, including data on species, sighting distance (perpendicular to the route), group size

Table 1  
Summary of data collection methodologies, 1983–1997

| Study area                                              | Census method                | Vehicle/horseback | Length/ (km)          | Number of data sets |
|---------------------------------------------------------|------------------------------|-------------------|-----------------------|---------------------|
| <i>A. Data collated from period prior to April 1997</i> |                              |                   |                       |                     |
| Gaysay                                                  | 1. Road count; 1983–1993     | V                 | 5.5–17.6 <sup>a</sup> | 88                  |
| Gojera                                                  | 1. Road count; 1986–1991     | V/H               | 15.4                  | 26                  |
| Upper Web Valley                                        | 1. Circuit census; 1988–1996 | H                 | 20.0                  | 23                  |
| Sanetti Plateau                                         | 1. Road count; 1983–1997     | V                 | 32.0                  | 109                 |
| <i>B. Data collected April to August 1997</i>           |                              |                   |                       |                     |
| Gaysay                                                  | 1. Road count                | V                 | 6.9–12.1 <sup>a</sup> | 2                   |
|                                                         | 2. Transects                 | H                 | 10.5                  | 8                   |
| Gojera                                                  | 1. Road count                | H                 | 8.6                   | 7                   |
| Upper Web Valley                                        | 1. Circuit census            | H                 | 20.0                  | 5                   |
|                                                         | 2. Transects                 | H                 | 30.0                  | 8                   |
| Sanetti Plateau                                         | 1. Road count                | V                 | 32.0                  | 2                   |

<sup>a</sup> The length of road counts in this area varied both seasonally, due to changing accessibility of the vehicle tracks, and over longer periods, due to changes in the de facto boundaries of the area.

and (for wildlife species) demographic composition, habitat type, and weather conditions; and (3) standardised transects were designated along East–West Universal Transverse Mercator lines in the Upper Web Valley and Gaysay, and were ridden on horseback. Sighting data were collected as for road counts. A summary of data collated and collected is given in Table 1.

### 2.5. Analysis

Where possible, census data were analysed both by finite population sampling (FPS) and by Distance Sampling (Buckland et al., 1993), the latter using “DISTANCE” software (Laake et al., 1994). Variances were large and consequently the low limit of the 95% confidence interval ( $DS_{min}$ ) was used as a conservative estimate of population sizes. For less frequently sighted species, other indices were also used. In particular, encounter rates (individuals of a species seen per kilometre traversed) have been used previously for analysis of census data from BMNP and may thus indicate relative changes in species’ abundances. Due to differences in the number of censuses conducted each year and in the variance in annual census results, long term data sets were analysed using the Kruskal–Wallis ANOVA by ranks and Dunn’s post-hoc test for unequal groups (Zar, 1996, eq. 11.28). Data collected more systematically were compared using 2-way ANOVA.

## 3. Results

A wide range of large mammals is found in the four study sites and sightings during the 1997 study period included a variety of antelope species (bushbuck, grey duiker, klipspringer, mountain nyala and reedbuck), several carnivores (Ethiopian wolf, golden jackal [*Canis aureus*], spotted hyaena [*Crocuta crocuta*] and serval [*Felis serval*]), two primate species (olive baboon [*Papio anubis*] and black and white colobus [*Colobus guereza guereza*]) and warthog (*Phacochoerus aethiopicus*), as

well as domestic animals including cattle and oxen, transport animals and small stock. Only the more frequently sighted species will be discussed in this section; the applicability of the census methodologies for inferring useful information on the less frequently sighted species will be discussed further in Section 4.1. Of the large mammals sighted in these study sites previously, only three were not seen during the 1997 study period. These were lion (*Panthera leo*) which were occasionally sighted during the 1980s (Hillman, 1986) but have not been seen over the past decade, and both leopard (*Panthera pardus*) and caracal (*Felis caracal*), sightings of which have been very infrequent during any period.

### 3.1. Large mammalian wildlife

Due to their conservation interest, monitoring efforts in BMNP have tended to emphasise the two endangered endemics, the mountain nyala and Ethiopian wolf. As a result, more information regarding these species is available, and they are discussed in greater detail below (Sections 3.2 and 3.3). Records of other censused species of large mammalian wildlife were largely restricted to the Gaysay and Gojera areas. Apart from mountain nyala, the most frequently sighted wildlife species in Gaysay were reedbuck, bushbuck, grey duiker and warthog (Fig. 2). For all three antelope, differences between years were very highly significant (Kruskal–Wallis,  $H_c > 31.264$ ,  $P < 0.001$ ) but Dunn’s post-hoc test lacked sufficient power to detect significant differences between successive years. However, comparisons between 1984 (the first year of regular censusing) and 1993 (the last year of censusing prior to the present study) showed highly significant declines (Dunn’s post-hoc test: bushbuck,  $Q = 4.157$ ,  $P < 0.005$ ; grey duiker,  $Q = 4.375$ ,  $P < 0.001$ ; reedbuck,  $Q = 4.099$ ,  $P < 0.005$ ). For warthog, however, although there were highly significant differences between years (Kruskal–Wallis,  $H_c > 24.725$ ,  $P < 0.01$ ), Dunn’s post-hoc test identified no significant differences between any pair of years and no trend is evident from the figure.

Analysis of transects conducted during 1997 indicated that all of these species were unevenly distributed throughout the Gaysay grasslands (Table 2). Differences between transect sessions were marginally significant only for reedbuck, thus data for this species were divided into sightings for May and June, and sightings for July and August. For the other species data were pooled across all four months for population analysis (Table 3).

Data from road counts in Gojera since 1986 show that this area was utilised by a variety of wildlife species (Table 4). Between 1986 and 1991 reedbuck, grey duiker and large troops of olive baboons were commonly sighted during road counts. During 1997 however, no reedbuck were sighted during seven road counts, whilst grey duiker and olive baboons were sighted in only low

Table 2  
Comparison of transects (two-way ANOVA; Gaysay, 1997)<sup>a</sup>

| Species     | Comparison of transects |     | Comparison of transect sessions |                 |
|-------------|-------------------------|-----|---------------------------------|-----------------|
|             | $F_{2,14}$              | $P$ | $F_{7,14}$                      | $P$             |
| Bushbuck    | 13.517                  | **  | 0.738                           | NS              |
| Grey duiker | 3.094                   | NS  | 0.716                           | NS              |
| Reedbuck    | 14.312                  | **  | 2.386                           | NS <sup>b</sup> |
| Warthog     | 6.945                   | *   | 1.004                           | NS              |

<sup>a</sup> NS, significant.

<sup>b</sup> Result marginally significant ( $P = 0.079$ ).

\*  $P \leq 0.01$ .

\*\*  $P \leq 0.001$ .

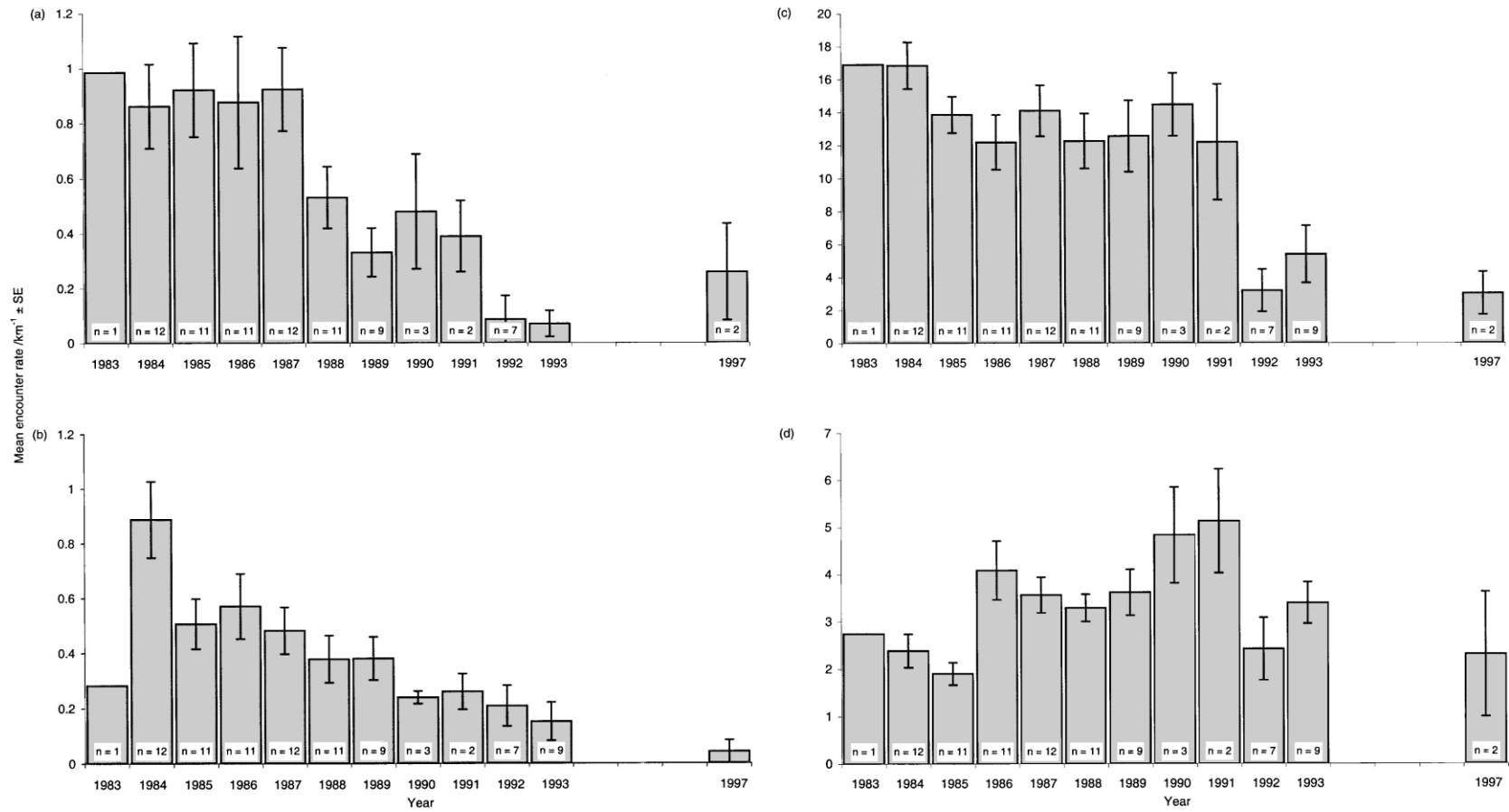


Fig. 2. Encounter rates for other regularly sighted wildlife during Gaysay road counts, 1983–1997. (a) Bushbuck, (b) grey duiker, (c) reedbuck, (d) warthog.

numbers. Spotted hyaena are largely nocturnal and sightings of this species have always been rare. Little can be inferred from the population estimates made for this species. The only species present in Gojera in comparable numbers in both 1986–1991 and 1997, were servals and warthog.

### 3.2. Mountain nyala

Between 1983 and 1997, mountain nyala were sighted in all of the study sites. Sightings in the Upper Web Valley have been infrequent throughout the period and rarely of more than a few individuals. Encounter rates for the other study sites are summarised in Fig. 3. For Gaysay, differences between years were very highly significant (Kruskal–Wallis,  $H_c > 31.264$ ,  $P < 0.001$ ) but as with the other antelope (Section 3.1) Dunn's post-hoc test lacked sufficient power to detect significant differences between successive years. However, comparisons between 1984 (the first year of regular censusing) and 1993 (the last year of censusing prior to the present study) showed a highly significant decline (Dunn's post-hoc test,  $Q = 4.479$ ,  $P < 0.001$ ). In Gojera there were also highly significant differences between encounter rates in different years (Kruskal–Wallis,  $H_c > 18.548$ ,  $P < 0.005$ ). Again, small sample sizes reduced statistical discrimination

but encounter rates were significantly lower in 1997 than in both 1987 and 1990 (Dunn's post-hoc test,  $Q > 3.038$ ,  $P < 0.05$ ). Data for the Sanetti Plateau were sparse and, as individual data points were not available prior to 1986, further statistical analysis was not conducted. Overall, however, there has been a tendency for sightings to decrease in all of the areas shown in Fig. 3 and the species is now sighted regularly only in Gaysay.

Distance data for mountain nyala were also examined. Data from road counts conducted between May and August of each year, from 1987 to 1990, showed no significant differences between years (two-way ANOVA,  $F_{2,6} = 0.138$ ,  $P > 0.05$ ) or between months ( $F_{3,6} = 3.505$ ,  $P > 0.05$ ). The data were pooled for analysis, giving estimates of 4600 (DS<sub>min</sub>) and 1700 (FPS).

There were no significant differences between numbers of mountain nyala sighted during different transect sessions from May to August 1997 and these data were pooled to give population estimates by both FPS and DS. These are given in Table 5 together with previous estimates, showing changes in the mountain nyala population in Gaysay.

A small group of nyala seen in the Upper Web Valley were probably members of a population reported from the Central Peaks and northern Sanetti Plateau area, and thought to number approximately 100–200. An

Table 3  
Population density estimates for Gaysay calculated from pooled results of all transects, 1997

| Species               | Sample size <sup>a</sup> |          | FPS estimate (95% CI) /km <sup>-2</sup> | DS estimate (95% CI) /km <sup>-2b</sup> |
|-----------------------|--------------------------|----------|-----------------------------------------|-----------------------------------------|
|                       | <i>n</i>                 | <i>c</i> |                                         |                                         |
| Bushbuck              | 49                       | 28       | 0.64 (–0.11–1.39)                       | –                                       |
| Grey duiker           | 31                       | 28       | 2.06 (0.26–3.86)                        | –                                       |
| Reedbuck <sup>c</sup> | 414                      | 96       | 10.80 (7.33–14.26)                      | 55.29 (37.11–82.36)                     |
| Reedbuck <sup>d</sup> | 162                      | 60       | 4.22 (–0.68–9.13)                       | 34.44 (18.83–62.99)                     |
| Warthog               | 285                      | 98       | 3.71 (0.37–7.06)                        | 13.14 (7.76–22.23)                      |

<sup>a</sup> *n*, total individuals seen; *c*, number of clusters.

<sup>b</sup> DS conducted only where  $c \geq 40$ .

<sup>c</sup> Population density estimate based on a data from May–June.

<sup>d</sup> Population density estimate based on a data from July–August.

Table 4  
Large mammal population estimates (by FPS) for Gojera, based on road count data, 1986–1997

| Species        | 1986–1991 (area = 41.0 km <sup>2</sup> )<br>estimate (95% CI) | 1997 (area = 30.7 km <sup>2</sup> )<br>estimate (95% CI) |
|----------------|---------------------------------------------------------------|----------------------------------------------------------|
| Bushbuck       | 2 (–5–10)                                                     | –                                                        |
| Grey duiker    | 21 (–59–51)                                                   | 2 (–58–12)                                               |
| Olive baboon   | 73 (–5234–380)                                                | 16 (–539–71)                                             |
| Reedbuck       | 35 (–553–123)                                                 | –                                                        |
| Serval         | 13 (–540–65)                                                  | 13 (–553–79)                                             |
| Spotted hyaena | 3 (–526–32)                                                   | –                                                        |
| Warthog        | 28 (–550–106)                                                 | 30 (–5101–160)                                           |

Table 5  
Approximate estimates of mountain nyala populations in Gaysay/Gasuray, 1983–1997

| Period:                                | 1983–1985 <sup>a</sup> | 1987–1990 <sup>b</sup> | 1991–1994 <sup>c</sup> | 1997 <sup>d</sup>  |
|----------------------------------------|------------------------|------------------------|------------------------|--------------------|
| Protected habitat available:           | 51 km <sup>2</sup>     | 106 km <sup>2</sup>    | 48 km <sup>2</sup>     | 51 km <sup>2</sup> |
| FPS population estimate:               | 1100                   | 1700                   | 150                    | 530                |
| DS <sub>min</sub> population estimate: | –                      | 4600                   | –                      | 1000               |

<sup>a</sup> Hillman (1986).

<sup>b</sup> 1997 analysis of previous road count data.

<sup>c</sup> Woldegebriel (1996).

<sup>d</sup> Transects, this study, 1997.

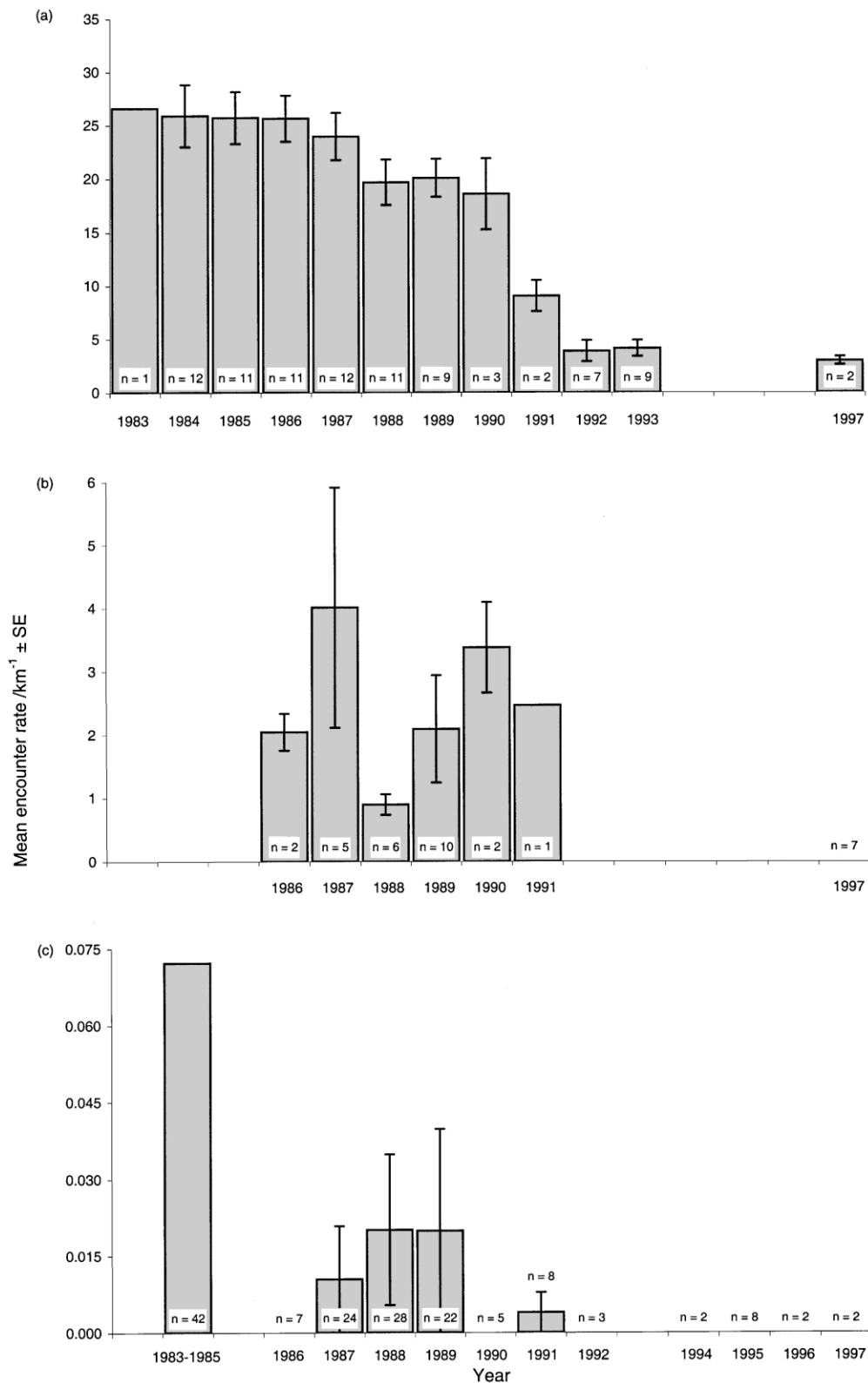


Fig. 3. Encounter rates of mountain nyala during routine road count censusing, 1983–1997. (a) Gaysay (no counts between 1994 and 1996); (b) Gojera (1986–1991 and 1997 only); (c) Sanetti Plateau (no counts in 1993), data for 1983–1985 from Hillman (1986) — no standard error available. *n* = numbers of censuses in each year.

additional 60 to 100 nyala occur at high density in a small patch of fenced woodland around the BMNP headquarters in Dinsho. More intensive work is required to assess the status of mountain nyala but the available information implies a total population of between 1100 and 1300 in BMNP.

### 3.3. *Ethiopian wolf*

As with mountain nyala, Ethiopian wolves were sighted in all study sites between 1983 and 1997. In Gaysay, sightings during road counts have been infrequent and only once have more than two wolves been sighted during a single count (four wolves sighted in October 1992). It is likely that a single pack or pair of wolves has been resident in Gaysay throughout the period for which data exist. Fluctuations in numbers of sightings

might reflect no more than reproductive success in a given year.

In Gojera, between 1987 and 1990, from one to six wolves (mean  $1.8 \pm 1.9$  S.D.) were sighted during 17 out of 22 censuses. By contrast, only one wolf was sighted during the seven censuses in 1997. Together with more detailed behavioural observations, these data indicate that whilst Gojera provided habitat for a pack of wolves during the late 1980s, no wolves are now permanently resident in the area.

For the Upper Web Valley and Sanetti Plateau, encounter rates during routine censusing are summarised in Fig. 4. For the Upper Web Valley differences between years were highly significant (Kruskal–Wallis,  $H_c > 19.019$ ,  $P < 0.005$ ) but Dunn's post-hoc test lacked sufficient power to detect significant differences between successive years. However, a comparison between 1989

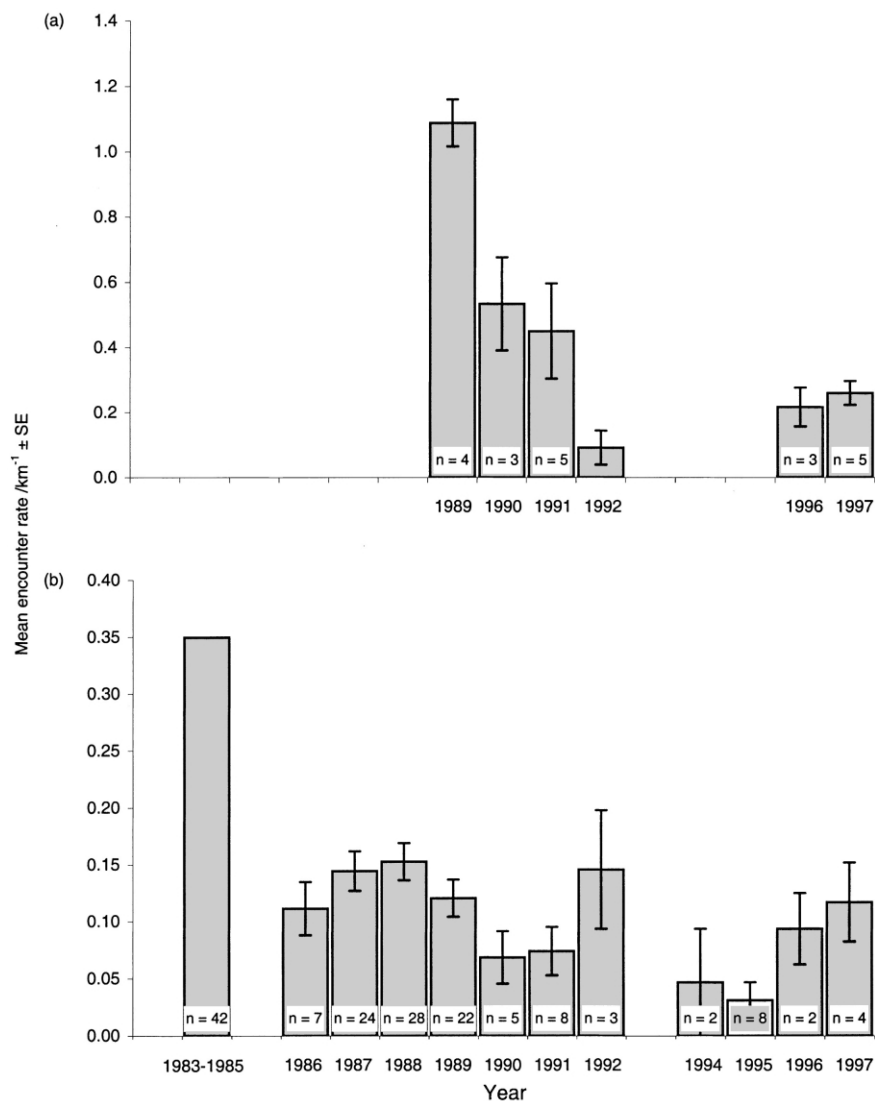


Fig. 4. Encounter rates of Ethiopian wolves during routine censusing, 1983–1997. (a) Upper Web Valley (no counts between 1993 and 1995); (b) Sanetti Plateau (no counts during 1993), data for 1983–1985 from Hillman (1986) — no standard error available.  $n$  = numbers of censuses in each year.

and 1992 showed a highly significant decline (Dunn's post-hoc test,  $Q = 3.827$ ,  $P < 0.002$ ). Similarly, although individual data points were not available for the Sanetti Plateau prior to 1986, for 1986–1997 differences between years were highly significant (Kruskal–Wallis,  $H_c > 23.209$ ,  $P < 0.01$ ). Encounter rates during 1995, the year during which sightings were lowest, were significantly lower than in 1987 and 1988 (Dunn's post-hoc test,  $Q = 3.621$ ,  $P < 0.02$  and  $Q = 3.815$ ,  $P < 0.01$ , respectively). Indications from 1996 and 1997 are that numbers are increasing again, following the low encounter rates of 1994 and 1995. However, numbers still appear to be well below those present during the period 1983–1985. Overall, encounter rates declined in both areas towards the early or mid-1990s but appear to be recovering. However, in neither area are sighting rates close to their previous maxima.

In the Upper Web Valley, wolves were encountered at an average of  $0.37 \text{ km}^{-1}$ , during transects carried out in 1997. There were marked differences between the six transects (two-way ANOVA;  $F_{5,35} = 8.178$ ,  $P < 0.001$ ) but no significant difference between different sets of transects ( $F_{7,35} = 0.906$ ,  $P = 0.513$ ), allowing sightings of wolves from all sets of transects to be pooled. This gave an estimated density of wolves of  $1.08 \text{ km}^{-2}$  by DS (95% CI =  $0.73\text{--}1.59 \text{ km}^{-2}$ ) and  $0.37 \text{ km}^{-2}$  by FPS (95% CI =  $0.18\text{--}0.56 \text{ km}^{-2}$ ).

### 3.4. Livestock and settlement

Livestock were the most frequently sighted animals in all study sites except Gaysay and, over recent years, their numbers showed marked increases in all four study areas. In Gaysay, domestic stock were infrequently sighted between 1983 and 1993. Transects conducted during 1997 provided evidence of an increased presence of domestic stock in this area. The majority of livestock sightings were of cattle and transport animals which are regularly driven through the grasslands between Dinsho and other, smaller villages. However, there is also evidence that transport animals in particular, but also some cattle and small stock, are now regularly grazed in Gaysay. Both FPS and DS gave current estimates of cattle densities of approximately  $4.5 \text{ km}^{-2}$ , whilst estimates of the density of transport animals ranged from  $5.0 \text{ km}^{-2}$  (by FPS) to  $8.3 \text{ km}^{-2}$  (by DS<sub>min</sub>). Much of the increase in livestock numbers in Gaysay arises from the increased size of all settlements neighbouring these grasslands.

Seasonality in livestock numbers is pronounced in the Upper Web Valley, confounding comparisons between years in which censuses were conducted in different months. Nonetheless, circuit census data provide clear indications that pastoralist activity in the area is increasing. From 1988 to 1992 cattle were the most commonly sighted animals (maximum 3003), followed

by small stock (maximum 1195) and transport animals (242). Censuses conducted in 1997 revealed much higher figures, with mean wet season cattle numbers 75–100% larger than those for 1988–1992. The maximum number of cattle counted in 1997 was almost 50% higher than the previous maximum and more small stock and transport animals were counted than ever before. Estimates of livestock populations in the Web Valley from transects also showed that cattle were the most commonly sighted species. Estimates of peak cattle populations varied from almost 8500 (FPS) to 9500 (DS<sub>min</sub>). Transport animals were estimated to peak at almost 900 (FPS) to 1400 (DS<sub>min</sub>). There are now seven villages distributed around the edges of the Upper Web Valley. All house permanent residents, as well as additional herdsman and their families during the wet season. Considerably more houses are present in this area than were counted in 1989.

For both Gojera (Fig. 5) and the Sanetti Plateau (Fig. 6), comparisons of livestock numbers between years were possible. It is unclear how consistently numbers of transport animals have been recorded on Sanetti and consequently these are not given. However, for all other livestock in both study sites, there were highly significant differences between years (Kruskal–Wallis,  $H_c > 18.548$ ,  $P < 0.005$ ). The low number of censuses conducted in some years reduced statistical discrimination but for all livestock species in Gojera, Dunn's post-hoc test identified differences between numbers sighted during 1997 and numbers sighted during several previous years (see Fig. 5 for further detail). The same approach was used for cattle on Sanetti (see Fig. 6a), whilst for small stock, significant to highly significant differences (Dunn's post-hoc test,  $P < 0.05$  to  $< 0.001$ ) were found between all counts taken in 1995–1997 and earlier years. Overall, livestock numbers showed a large but steady increase on Sanetti from 1986 to 1997 but in Gojera the increase was considerably more abrupt. At least 76 dwellings were counted within the 1974 BMNP boundary of Gojera, and populations of approximately 4000 cattle, 1500 small stock and 700 transport animals were estimated (using DS<sub>min</sub>). During 1997, domestic dogs were sighted more often than any wildlife species in this study site.

## 4. Discussion

### 4.1. Appraisal of the methods

The long-term census records from BMNP make this area particularly important for insights into anthropogenic effects on wildlife. Unfortunately, BMNP suffers in common with many other protected areas in historically having lacked a scientifically robust monitoring scheme. A rigorous monitoring system was first

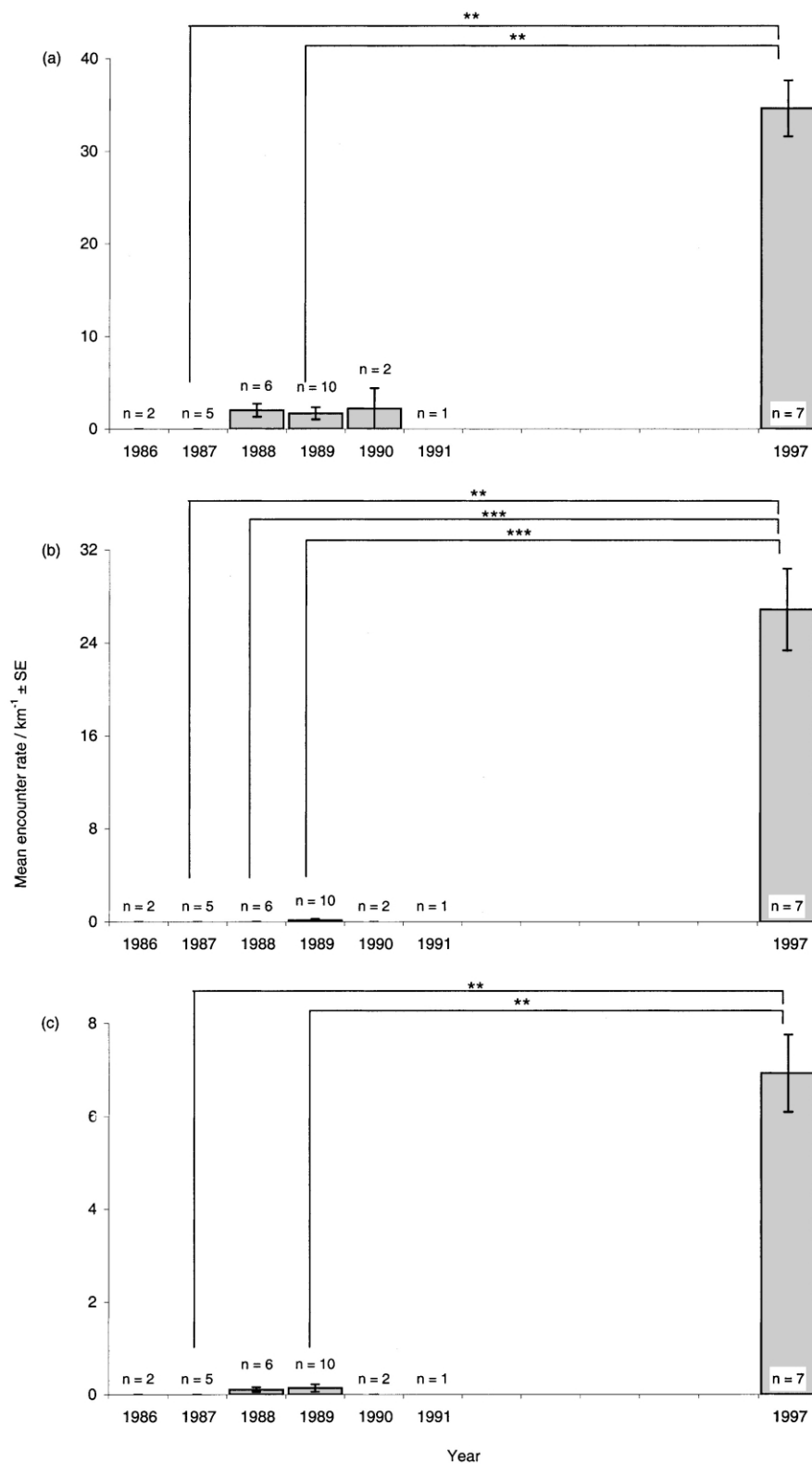


Fig. 5. Encounter rates of livestock in Gojera, 1986–1997. (a) Cattle, (b) small stock, (c) transport animals. Differences between years are marked on the figure where appropriate (Dunn's post-hoc test: \* significant,  $P < 0.05$ ; \*\* highly significant,  $P < 0.01$ ; \*\*\* very highly significant,  $P < 0.001$ ).

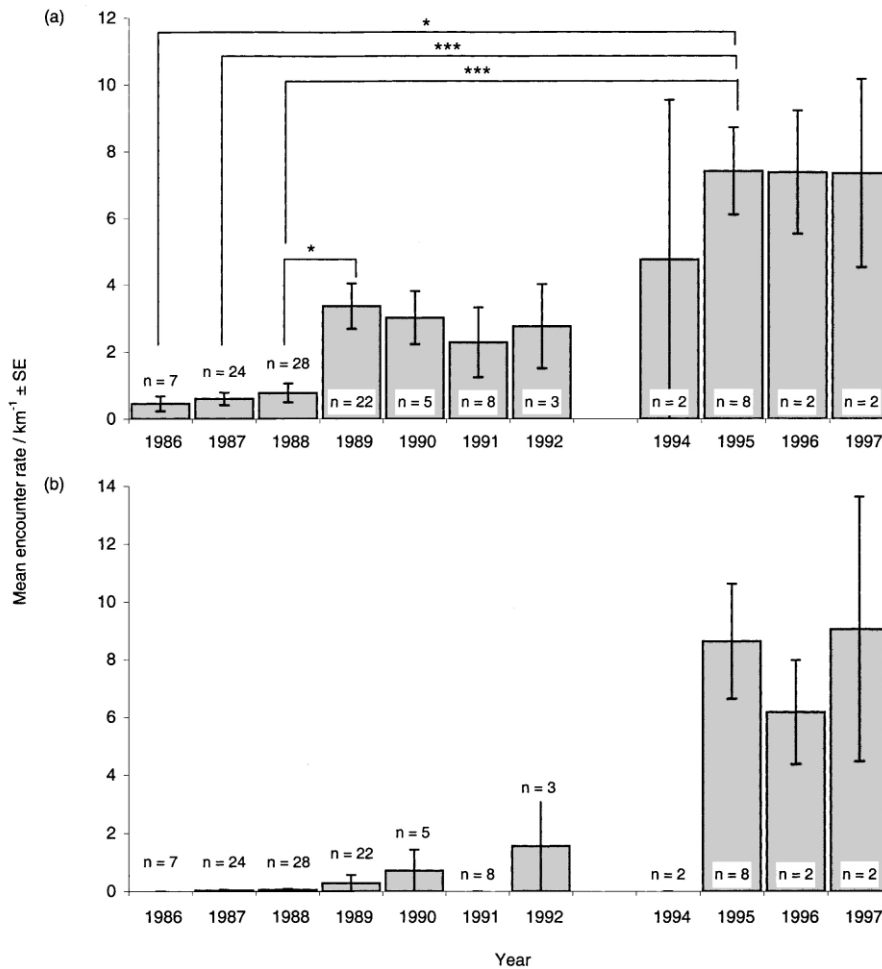


Fig. 6. Encounter rates of livestock on the Sanetti Plateau, 1986–1997. (a) Cattle, (b) small stock. Differences between years are marked on the figure where appropriate (Dunn's post-hoc test: \* significant,  $P < 0.05$ ; \*\* highly significant,  $P < 0.01$ ; \*\*\* very highly significant,  $P < 0.001$ , but see text for further details).

instituted during preparation of BMNP's management plan (Hillman, 1986) but has since been continued in a sporadic and inconsistent fashion. Much of the burden of censusing has fallen to visiting researchers who are, inevitably, absent from BMNP for prolonged periods. Despite these problems, the census records available for BMNP are the best available for any protected area in Ethiopia and allow simple assessments of trends in large mammal abundance. However, detailed comparative analysis of the earlier data with those collected for this study was complicated by inconsistencies in data collection techniques over time and, for certain species at least, by the need for more appropriate monitoring methodologies.

In the past, routine censusing has been used to estimate populations of all large mammal species in BMNP (Hillman, 1986). However, estimation of population densities by this method is possible only for more frequently sighted species. For the carnivores, including wolves (in Gaysay), golden jackals, spotted hyaenas and servals, sightings are very infrequent and only species-

specific investigations will provide accurate data on population trends. Olive baboons highlight the inadequacies of routine censusing for estimating populations of an infrequently sighted, group-living species, where large changes in populations may be erroneously inferred from small differences in the number of groups sighted.

For more frequently sighted species, analysis of population densities may employ either FPS or DS. Comparisons of these methods suggest that choice of technique may be critical. Assessment of species-specific sighting distances for use in FPS is complex and, where sample sizes are sufficient, DS is preferable. However, large variances in DS estimates resulted both from small sample sizes, and from significant differences in animal distributions across study sites. Stratification of sampling effort by habitat type (Buckland et al., 1993) will refine this technique for future use. The difference between cattle numbers sighted during the circuit census of the Upper Web Valley (maximum 4500) and estimates from transects (8000–9500 cattle, by FPS and

$DS_{min}$ , respectively), suggest that there may be problems with the transect design. The circuit census is close to a total count of the area and the inflated estimates from transects suggest that possible deviations involved in circumnavigating major obstacles (such as swamps) might have made the transects significantly longer than their direct linear distances. In Gaysay, analysis of previous (1987–1990) road count data gave an estimate of mountain nyala numbers by  $DS_{min}$  of between 2.5 and 3 times that obtained by FPS. It is likely that the road in Gaysay provides a fairly complete view over the entire area. Sighting distance distributions are therefore likely to reflect observer decisions on which point represents the best vantage from which to count a particular herd, for example, rather than arising as a result of natural probabilities of detection. This might explain elevated estimates from DS and suggest that for this area at least, DS should not be used to analyse road count data.

Population estimates derived by extrapolation of densities to larger areas should be treated with caution. Censuses in the Gaysay grasslands have traditionally been taken to be representative of the whole of Gaysay/Gojera. This approach may not be justified however, and the establishment of new censuses for the woodland areas would provide more accurate data on species' abundances.

Problems experienced in this study, including long-term inconsistencies in data collection methods, the importance of appropriate methodologies, and the need for stratified sampling, are by no means unique to monitoring in BMNP. Rather, these serve to highlight the need for consistency and continuity in methodologies, reinforcing the importance of scientifically rigorous monitoring schemes in the management of any protected area. Despite these problems, however, monitoring data from BMNP do allow a meaningful assessment of trends of abundance for a variety of species, both wild and domestic, and these are discussed in the following section.

#### 4.2. *Wildlife, livestock and settlement*

The results of this study show major changes in the abundances of a variety of wildlife species, together with increases in livestock and dogs that pose a serious threat to the conservation of BMNP. Of particular concern to conservationists, are declines in populations of the mountain nyala and Ethiopian wolf, which are discussed briefly, before moving on to other species and broader conservation issues.

##### 4.2.1. *Mountain nyala and Ethiopian wolf*

Mountain nyala have shown an overall decline in BMNP over the period 1987–1997. Our results suggest a 1997 population of between 1100 and 1300 mountain nyala, considerably fewer than during 1987–1990.

However, since 1991 there has been an encouraging indication of a reduction in levels of persecution and of an incipient population recovery in Gaysay, the most important area for this species. The lack of nyala sightings during road counts on Sanetti Plateau since 1991, might be attributable to increasing settlement and livestock grazing, as well as concurrent poaching. However, more regular censuses and/or a detailed study of the nyala population are required to determine diurnal movement patterns and population fluctuations in this area.

The Ethiopian wolf has also suffered major declines in BMNP, particularly in the Upper Web Valley and Sanetti Plateau, the two most important areas for this species. Although sighting rates appear indicative of recent increases in both areas, recovery is still in its early stages following a serious decline, and many threats remain. In Gaysay, numbers of wolves seen during censuses were low, restricted to members of one known pack. Tall vegetation hinders hunting of rodents by wolves (Malcolm, 1976 cited in Malcolm and Sillero-Zubiri, 1997) and improved enforcement of anti-grazing regulations in Gaysay may thus have a negative effect on utilisation of the area by wolves. Shrub vegetation was reportedly taller in the 1997 wet season than in previous years (E. Ebu, personal communication). Measures to reduce vegetation cover and encourage wolves to use the area may well be to the detriment of the mountain nyala, presenting a conservation dilemma. However, Gaysay is relatively more important for the mountain nyala than for the wolf and it may be that the habitat requirements of nyala should be placed above those of the wolf in this area.

Wolf sighting rates during regular censuses of the Upper Web Valley and Sanetti Plateau correlate well with more intensive demographic field observations. Declines during the late 1980s and early 1990s are principally attributable to outbreaks of disease in both Sanetti Plateau and the Upper Web Valley (Sillero-Zubiri, 1994). An outbreak of rabies occurred between November 1991 and February 1992 in the Upper Web Valley, as a result of which 77% of wolves from five known packs died or disappeared (Sillero-Zubiri et al., 1996b). There was also direct persecution of wolves by humans in the wake of the change of government (Sillero-Zubiri and Gottelli, 1993). Numbers of wolves sighted are still lower than in 1989–1991 but have shown an upward trend since the mid-1990s, giving grounds for cautious optimism. Recovery is slow, as would be expected given the low number of breeding units (breeding pairs, with or without auxiliary pack members) which remained in 1992 (Sillero-Zubiri, 1994).

The Upper Web Valley was the only area where current wolf populations could be reliably assessed and an estimate of 50 wolves was calculated (using  $DS_{min}$ ). There have been several previous attempts to estimate

the wolf population of this area. The most reliable of these come from work conducted since the late 1980s. At that time, densities of wolves in the Upper Web Valley were given as  $1.0\text{--}1.2\text{ km}^{-2}$  (Gottelli and Sillero-Zubiri, 1992), nearly double the estimates from the current study. In 1994, nine wolves were seen in 140 km of transects in the Web Valley area — an encounter rate of  $0.06\text{ km}^{-1}$  (J. Malcolm, personal communication). In 1997 wolves were encountered at a rate of  $0.37\text{ km}^{-1}$ . Such an increase over three years is encouraging, supporting other data from the Upper Web Valley and Sanetti Plateau.

#### 4.2.2. *Other large mammalian wildlife*

Other species of large mammalian wildlife were seen only infrequently in the Upper Web Valley and Sanetti Plateau during 1997, whilst large mammalian wildlife was almost eradicated from Gojera. However, 1997 censuses in Gaysay produced estimates of wild ungulates and warthog that could be compared with earlier data. Encounter rates of reedbuck, bushbuck and duiker changed significantly from 1984 to 1993, showing similar downward trends to nyala. Data for these species are too limited for the assessment of trends since 1991. However, current populations in Gaysay are estimated at 520 reedbuck (using  $DS_{\min}$ ) and approximately 100 duiker (by FPS). The reedbuck estimate is based on data from May and June 1997; due to the tendency of reedbuck to lie up in long grass during the wet season (Hillman, 1986), sightings were limited during later months. Bushbuck are mainly found in the northern woodlands and will only be censused accurately if a census is introduced in this habitat. The warthog population in Gaysay was estimated at 400 (using  $DS_{\min}$ ). It is likely that densities of this species are higher in the woodlands than those measured in the grasslands however, and the population has therefore been underestimated. Evidence from both Gaysay and Gojera, suggest that warthog were the only commonly sighted species that appeared largely unaffected by changes in livestock and settlement within BMNP.

#### 4.2.3. *Livestock and settlement*

In contrast to the results for almost all censused wildlife, livestock numbers and settlement have increased in all study areas since 1983. The most striking case was Gojera, where intensive cultivation and other usage by humans and livestock, and minimal presence of wildlife, renders the area largely worthless as a wildlife corridor. The most abrupt increase in livestock numbers occurred between the late 1980s and 1997, and is probably associated with resettlement of this area following the change of government in 1991. Similar, but less abrupt increases are evident in all other study sites during the same period.

Increasing settlement and livestock numbers are major concerns for three reasons. First, the erosive

effects of livestock are already clearly visible in many areas. The detrimental effects of this on the natural flora and fauna of BMNP have been discussed elsewhere (Hillman, 1988; Woldegebriel, 1996). Second, high numbers of livestock have a negative impact on tourists, particularly in the Upper Web Valley. This area is one of the greatest attractions for tourists to BMNP yet their experience is greatly impaired by the presence of thousands of livestock. Similarly, the Gaysay grasslands situated on the main Dinsho-Adaba road, are a showcase for BMNP. The presence of domestic stock in the area has a very negative visual impact. Third, increased settlement and numbers of livestock are linked to a variety of threats to the wildlife of BMNP, notably to the mountain nyala and Ethiopian wolf. From a wildlife viewpoint, pastoralism is a form of interference competition (Happold, 1995). Nyala and other antelope compete directly with livestock for food and no nyala are seen in areas where livestock pressure is high (Brown, 1969). Domestic dogs, kept to protect livestock from hyaenas, are also a direct threat to nyala and other antelope. Most households keep one or two dogs that are inadequately provisioned and only loosely controlled. Many are free-ranging and are known to kill nyala calves and juveniles each year; they have also been observed chasing adult nyala.

The results of human-induced disturbance of Ethiopian wolf populations are more complex. There is evidence to suggest that cattle compete with rodents (Delaney, 1972) and would thus reduce the prey-base of wolves. In a long-term exclusion experiment in Kenya, the exclusion of large ungulates led to an increase of 60% in the populations of small mammals (Keesing, 1998, 2000). Conversely, it has been suggested both that wolves use cattle as a “mobile hide” whilst foraging, and thereby increase their hunting success, and that in keeping pasture short, cattle actually encourage rodent communities in the area, promoting wolf prey biomass (Sillero-Zubiri and Gottelli, 1995). More detailed studies are required to assess the interactions between livestock, small mammals and Ethiopian wolves in this area.

The clearest threats to Ethiopian wolves are those posed by domestic dogs. Dogs compete with wolves for prey (Sillero-Zubiri, 1994), spread disease (Sillero-Zubiri et al., 1996b) and hybridise with wolves (Gottelli et al., 1994). It is evident that disease is one of the major agents of decline of the wolves. Dogs act as reservoirs for a number of canid-related diseases including rabies, canine distemper and parvovirus. It is perhaps significant that the first evidence of an increase in wolf numbers in BMNP since 1989 follows the inception (in 1995) of a major vaccination programme for domestic dogs in the Bale Mountains (Laurenson et al., 1997).

Concerns over hybridisation were sustained by the only mating observed during the study period—between

a female wolf and a male dog. Although the wolves live in packs with a dominant breeding pair, a high proportion of matings are between the dominant female and a male from outside her pack (Sillero-Zubiri et al., 1996a). Higher numbers of free-ranging dogs living sympatrically with the wolves will increase the probability of extra-pack copulations between wolves and dogs. An average of 10 dogs free-ranging in the Upper Web Valley area each day was estimated (by FPS). As with other FPS estimates, this may well be an underestimate.

#### 4.3. Management recommendations

Recommendations arise from this study relating to two main issues in the future management of BMNP, comprising monitoring and reducing negative human impacts.

##### 4.3.1. Monitoring

The study highlights the need to continue with a consistent approach to census methodologies, while also seeking to adopt more precise methods and to fill important gaps in areas censused.

Census protocols developed across four study areas in the 1980s provide simple but informative indices of population abundance for a variety of large mammal species. These could be used regularly and consistently, to continue to provide managers with a long-term view on population trends of large mammals on BMNP grasslands.

In addition, transect censusing, refined by stratification of sampling effort and conducted less frequently, will provide better estimates of absolute population sizes for all censused species. Further work is required to assess effective sighting distances for use with FPS, whilst species-specific studies may be required to provide information on BMNP's less frequently sighted mammals.

The woodland areas of Gaysay are not well covered, and there are no accurate data on a variety of species that are less often sighted in the grasslands, particularly Menelik's bushbuck. A census protocol could be introduced to this area, which would also provide useful information on the extent of livestock grazing in the woodland areas.

The task of routine monitoring in BMNP should not fall to visiting researchers, but rather be incorporated as a key component within the work plan of a BMNP biologist. In this regard, the lack of a full-time staff biologist since 1992 is disappointing and consideration could be given by the Park authorities to re-instating this post.

##### 4.3.2. Reducing human impact

The study has shown that increasing human impact, including settlement and grazing of livestock, has had a

serious impact on the wildlife living in BMNP's grasslands, including two key endemic species, the mountain nyala and the Ethiopian wolf that the park was established to safeguard. In the short-term, continued investment in community education and domestic dog vaccination programmes may alleviate some of the immediate causes of concern for the Ethiopian wolf and other wildlife threatened by irresponsible dog ownership. Beyond this, however, serious consideration could be given to more fundamental aspects of BMNP's management.

The generally accepted definition of a National Park established in IUCN's Category II, such as BMNP, does not allow for settlement or livestock within the area (IUCN, 1998). The management recommendations of Hillman (1986) were formulated some 14 years ago as BMNP was established, and may require urgent reconsideration in light of the current status of BMNP's wildlife. Besides establishing a programme of monitoring and research, Hillman's recommendations included:

- (1) more extensive Park management;
- (2) a resettlement programme;
- (3) more integrated rural development;
- (4) changes to the BMNP boundaries; and,
- (5) the introduction of a 5km wide buffer zone.

Due mostly to the disruptions and setbacks caused during the period of unrest in 1991, none of these recommendations has been implemented. Monitoring and management of BMNP are now largely confined to Gaysay and areas easily accessible from the headquarters at Dinsho. In order to protect and manage the array of other habitats and species in BMNP, consideration could be given by the Park authorities to renewed investment in providing staff outposts for the effective protection of other areas of BMNP. Whether wholesale resettlement of people to areas outside BMNP is now possible, consideration could be given to preventing further settlement in BMNP and to reducing current levels of habitation and utilisation.

Although there are positive signs of the recovery of BMNP's wildlife in recent years, only by a reduction in human disturbance in the area will further degradation of the area be prevented and the security of resident wildlife increased. In this regard, it is recommended that a consultative park planning exercise is initiated as soon as possible to re-visit Hillman's (1986) recommendations, and to establish BMNP management on a firmer footing than hitherto.

#### Acknowledgements

We are grateful to the Ethiopian Wildlife Conservation Organisation for permission to work in BMNP. Work was partly funded by the Durrell Institute for

Conservation and Ecology. The Ethiopian Wolf Conservation Programme (EWCP), supported by the Born Free Foundation, provided accommodation and field equipment. We are indebted to Edriss Ebu, Alo Hussein and Girma Tadesse for assistance in the field, and to BMNP Warden Fikadu Garedew and the EWCP, for access to results of past censuses. Thanks to Brian Davis, Rob Freckleton, James Malcolm, Jorgelina Marino, Bill Sutherland and two anonymous referees for useful criticisms and discussion.

## References

- Brown, L.H., 1969. Observations on the status, habitat and behaviour of the mountain nyala *Tragelaphus buxtoni* in Ethiopia. *Mammalia* 33, 545–597.
- Buckland, S.T., Anderson, D.R., Burnham, K.P., Laake, J.L., 1993. Distance Sampling: Estimating Abundance of Biological Populations. Chapman and Hall, London.
- Delaney, M.J., 1972. The ecology of small rodents in tropical Africa. *Mammal Review* 2, 1–42.
- Gottelli, D., Sillero-Zubiri, C., 1992. The Ethiopian wolf — an endangered endemic canid. *Oryx* 26, 205–214.
- Gottelli, D., Sillero-Zubiri, C., Applebaum, G.D., Girman, D., Roy, M., Garcia-Moreno, J., Ostrander, E., Wayne, R.K., 1994. Molecular genetics of the most endangered canid: the Ethiopian wolf, *Canis simensis*. *Molecular Ecology* 3, 301–312.
- Happold, D.C.D., 1995. The Interactions between humans and mammals in Africa in relation to conservation: a review. *Biodiversity and Conservation* 4, 395–414.
- Hillman, J.C., 1986. Bale Mountains National Park Management Plan, Report to Ethiopian Wildlife Conservation Organisation, Addis Ababa.
- Hillman, J.C., 1988. The Bale Mountains National Park area, south-east Ethiopia, and its management. *Mountain Research and Development* 8, 253–258.
- Hillman, J.C., 1993. Ethiopia: Compendium of Wildlife Conservation Information, Report to Ethiopian Wildlife Conservation Organisation, Addis Ababa.
- IUCN, 1998. United Nations List of Protected Areas. IUCN, Gland, Switzerland.
- Keesing, F., 1998. Impacts of ungulates on the demography and diversity of small mammals in central Kenya. *Oecologia* 116, 381–389.
- Keesing, F., 2000. Cryptic consumers and the ecology of an African savanna. *Bioscience* 50, 205–215.
- Laake, J.L., Buckland, S.T., Anderson, D.R., Burnham, K.P., 1994. DISTANCE User's Guide V2.1. Colorado State University, Fort Collins.
- Laurenson, M.K., Fekadu, S., Sillero-Zubiri, C., 1997. Disease, domestic dogs, and the Ethiopian wolf: the current situation. In: Sillero-Zubiri, C., Macdonald, D.W. (Eds.), The Ethiopian Wolf — Status Survey and Conservation Action Plan. IUCN, Gland, Switzerland, pp. 32–42.
- Malcolm, J.R., Sillero-Zubiri, C., 1997. The Ethiopian wolf: distribution and population status. In: Sillero-Zubiri, C., Macdonald, D.W. (Eds.), The Ethiopian Wolf — Status Survey and Conservation Action Plan. IUCN, Gland, pp. 12–25.
- Sillero-Zubiri, C., 1994. Behavioural Ecology of the Ethiopian Wolf. D. Phil. thesis, University of Oxford.
- Sillero-Zubiri, C., Gottelli, D., 1993. The plight of the Ethiopian wolf. *Canid News* 1, 10–11.
- Sillero-Zubiri, C., Gottelli, D., 1995. Diet and feeding behavior of Ethiopian wolves (*Canis simensis*). *Journal of Mammalogy* 76, 531–541.
- Sillero-Zubiri, C., Tattersall, F.H., Macdonald, D.W., 1995a. Bale Mountains rodent communities and their relevance to the Ethiopian wolf (*Canis simensis*). *African Journal of Ecology* 33, 301–320.
- Sillero-Zubiri, C., Tattersall, F.H., Macdonald, D.W., 1995b. Habitat selection and daily activity of giant molerats *Tachyoryctes macrocephalus*: significance to the Ethiopian wolf *Canis simensis* in the Afroalpine ecosystem. *Biological Conservation* 72, 77–84.
- Sillero-Zubiri, C., Gottelli, D., Macdonald, D.W., 1996a. Male philopatry, extra pack copulations and inbreeding avoidance in Ethiopian wolves (*Canis simensis*). *Behavioral Ecology and Sociobiology* 38, 331–340.
- Sillero-Zubiri, C., King, A.A., Macdonald, D.W., 1996b. Rabies and mortality in Ethiopian wolves (*Canis simensis*). *Journal of Wildlife Diseases* 32, 80–86.
- Sillero-Zubiri, C., Johnson, P.J., Macdonald, D.W., 1998. A hypothesis for breeding synchrony in Ethiopian wolves (*Canis simensis*). *Journal of Mammalogy* 79, 853–858.
- Woldegebriel, G.K., 1996. The status of mountain nyala (*Tragelaphus buxtoni*) in Bale Mountains National Park 1986–1994. *Walia* 17, 27–37.
- Wynne, G., 1998. Conservation policy and politics. In: Sutherland, W.J. (Ed.), Conservation, Science and Action. Blackwell Science, Oxford, pp. 256–285.
- Yalden, D.W., 1983. The extent of high ground in Ethiopia compared to the rest of Africa. *Sinet* 6, 35–38.
- Zar, J.H., 1996. Biostatistical Analysis, Prentice Hall, Upper Saddle River, New Jersey.